

③ & ④ can be used to change the transition voltage by tuning R_1 and R_2 . They are building block of schmitt trigger

Schmitt Triggers

→ Comparator with hysteresis

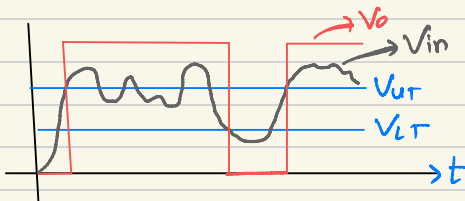
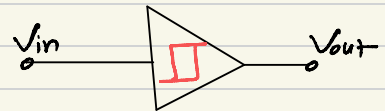
→ has two threshold voltages (References)

→ V_{UT} → upper threshold voltage.
→ V_{LT} → Lower " "

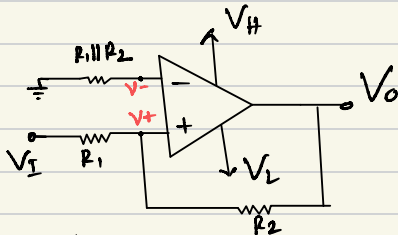
Non-Inverting Schmitt trigger:

→ V_{UT} → low to high transition.

→ V_{LT} → high to low transition.



⊛ Non-Inv. S.T. Ckt



$$\begin{cases} V_- = 0 \\ V_+ = V_o \frac{R_1}{R_1 + R_2} + V_I \frac{R_2}{R_1 + R_2} \end{cases}$$

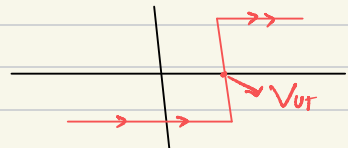
when $V_+ < V_- \rightarrow V_o = V_L$

$$\therefore V_+ = V_L \frac{R_1}{R_1 + R_2} + V_I \frac{R_2}{R_1 + R_2} < 0$$

$$\therefore V_I < \left(-\frac{R_1}{R_2} \right) V_L$$

$$V_o = V_L$$

$$\therefore V_{UT} = -\frac{R_1}{R_2} V_L$$



When $V_+ > V_- \rightarrow V_0 = V_H$

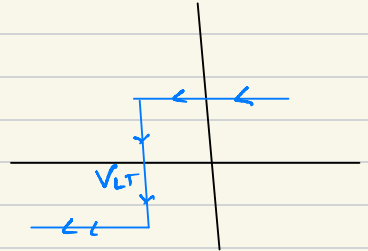
$$\therefore V_- = 0$$

$$V_+ = V_H \frac{R_1}{R_1 + R_2} + V_I \frac{R_2}{R_1 + R_2}$$

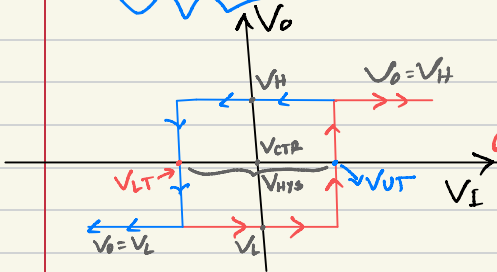
$$V_+ > V_-$$

$$\Rightarrow \boxed{V_I > \left(-\frac{R_1}{R_2}\right) V_H} \rightarrow V_0 = V_H$$

$$\therefore \boxed{V_{LT} = \left(-\frac{R_1}{R_2}\right) V_H}$$



Combined characteristics



*** $\therefore$ Formulas

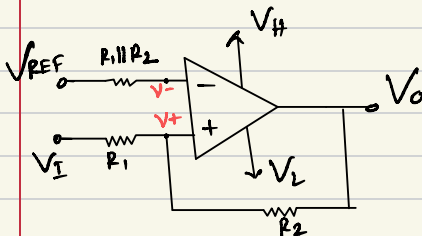
$$V_{UT} = \left(-\frac{R_1}{R_2}\right) V_L$$

$$V_{LT} = \left(-\frac{R_1}{R_2}\right) V_H$$

$$V_{HYS} = V_{UT} - V_{LT} \rightarrow \text{Hysteresis width}$$

$$V_{CTR} = \frac{V_{UT} + V_{LT}}{2} \rightarrow \text{Center voltage}$$

Non. Inv. S.T. with applied voltage. (V_{REF})



$$V_- = V_{REF}$$

$$V_+ = V_0 \frac{R_1}{R_1 + R_2} + V_I \frac{R_2}{R_1 + R_2}$$

$$V_{UT} = \left(-\frac{R_1}{R_2}\right) V_L + \left(\frac{R_1 + R_2}{R_2}\right) V_{REF}$$

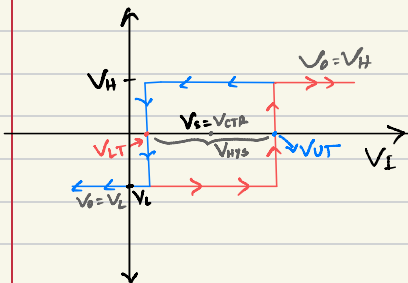
$$V_{LT} = \left(-\frac{R_1}{R_2}\right) V_H + \left(\frac{R_1 + R_2}{R_2}\right) V_{REF}$$

$$V_{HYS} = V_{UT} - V_{LT}$$

$$V_{CTR} = \frac{V_{UT} + V_{LT}}{2} = V_S$$

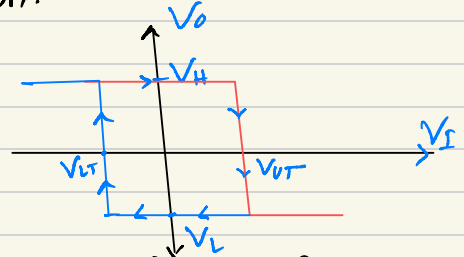
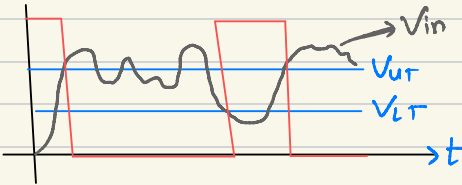
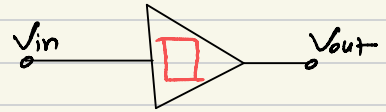
shift voltage, V_S

$$V_S = \frac{R_1 + R_2}{R_2} V_{REF}$$



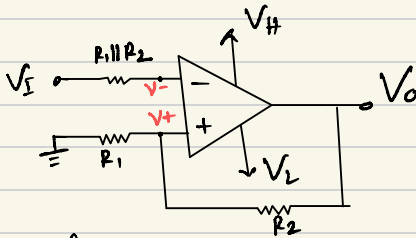
Inverting Schmitt triggers

$\rightarrow V_{UT} \rightarrow$ high to low transition.
 $\rightarrow V_{LT} \rightarrow$ low to high transition.



Transfer char. of Inv S.T.

(*) Inv. S.T. CKT



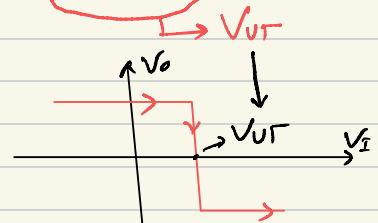
$$\begin{cases}
 V_- = V_i \\
 V_+ = V_o \frac{R_1}{R_1 + R_2}
 \end{cases}$$

when $V_- < V_+ \rightarrow V_o = V_H$

$$\begin{aligned}
 V_- &= V_i \\
 V_+ &= V_H \frac{R_1}{R_1 + R_2}
 \end{aligned}$$

$$V_- < V_+$$

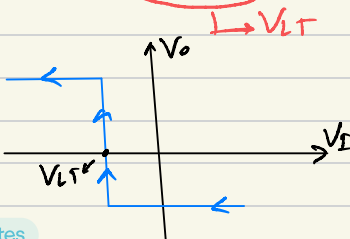
$$V_i < V_H \frac{R_1}{R_1 + R_2} \rightarrow V_o = V_H$$



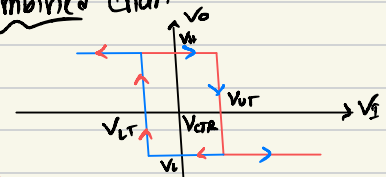
When $V_- > V_+ \rightarrow V_o = V_L$

$$\begin{aligned}
 V_- &= V_i \\
 V_+ &= V_L \frac{R_1}{R_1 + R_2}
 \end{aligned}$$

$$V_- > V_+ \rightarrow V_i > V_L \frac{R_1}{R_1 + R_2} \rightarrow V_o = V_L$$



Combined char.



** Formulas

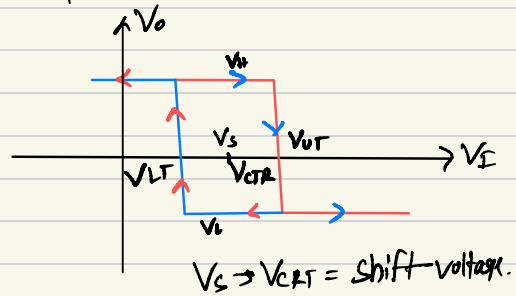
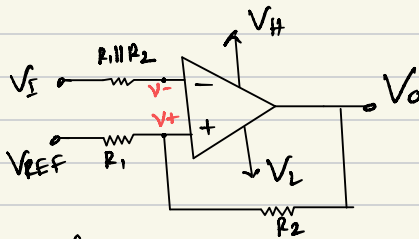
$$V_{UT} = V_H \frac{R_1}{R_1 + R_2}$$

$$V_{LT} = V_L \frac{R_1}{R_1 + R_2}$$

$$V_{HYS} = V_{UT} - V_{LT}$$

$$V_{CTR} = \frac{V_{UT} + V_{LT}}{2}$$

⊛ Inv. S.T. CKT with applied voltage (V_{REF})



$$\left\{ \begin{array}{l} V_- = V_I \\ V_+ = V_O \frac{R_1}{R_1 + R_2} + V_{REF} \frac{R_2}{R_1 + R_2} \end{array} \right.$$

** Formulas

$$V_{UT} = V_H \frac{R_1}{R_1 + R_2} + V_{REF} \frac{R_2}{R_1 + R_2}$$

$$V_{LT} = V_L \frac{R_1}{R_1 + R_2} + V_{REF} \frac{R_2}{R_1 + R_2}$$

$$V_{HYS} = V_{UT} - V_{LT}$$

$$V_S = V_{CTR} = \frac{V_{UT} + V_{LT}}{2} = V_{REF} \frac{R_2}{R_1 + R_2}$$

Practice problem

$$R_1 = 10k\Omega, R_2 = 90k\Omega, V_H = 10V, V_L = -10V$$

i) Determine the followings for non-inverting schmitt trigger.

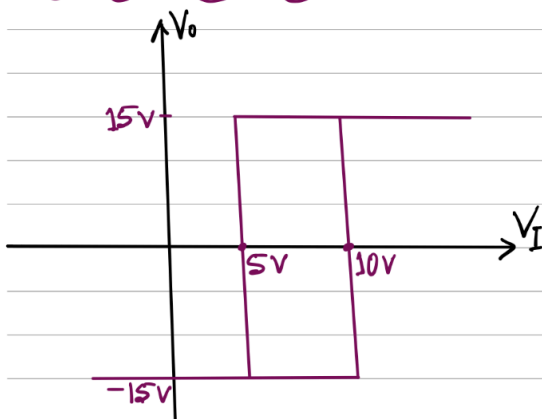
$$\rightarrow V_{UT}, V_{LT}, V_{HYS}, V_{CTR}$$

ii) Determine the values for inverting schmitt trigger.

⊛ Schmitt Trigger → Bistable multivibrator

because it has two stable states → $V_O = V_H$ & V_L

Practice Problem



From the given VTC, answer the following questions -

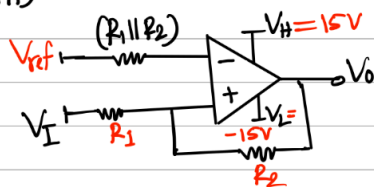
- i) Identify the Schmitt trigger.
- ii) Find, V_{Hys} , V_S , V_H , V_L , V_{UT} , V_{LT}
- iii) Design the S.T. ckt.
- iv) if $V_i = 20\sin 400\pi t$. Draw the V_o vs t graph.

Soln:

i) Non-Inv. Schmitt Trigger with applied voltage.

ii) $V_{Hys} = 10 - 5 = 5V$, $V_S = V_{CTR} = \frac{10+5}{2} = 7.5V$
 $V_H = +15V$, $V_L = -15V$, $V_{UT} = 10V$, $V_{LT} = 5V$

iii)



Now, we need to find out R_1 , R_2 & V_{ref}

$$V_{UT} = -\frac{R_1}{R_2} V_L + \frac{R_1 + R_2}{R_2} V_{ref} \quad \text{--- (i)}$$

$$V_{LT} = -\frac{R_1}{R_2} V_H + \frac{R_1 + R_2}{R_2} V_{ref} \quad \text{--- (ii)}$$

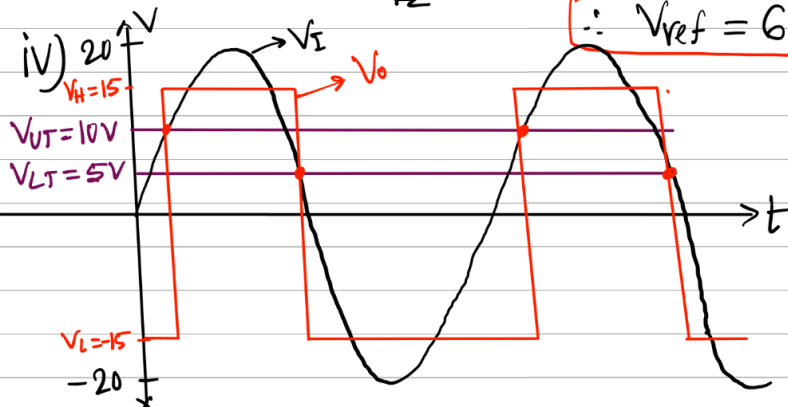
$$(i) - (ii) \quad V_{UT} - V_{LT} = 5 = \frac{R_1}{R_2} (V_H - V_L)$$

$$\Rightarrow 5 = \frac{R_1}{R_2} 30 \quad \therefore \frac{R_1}{R_2} = \frac{1}{6}$$

Let, $R_1 = 10K \therefore R_2 = 60K\Omega$

Now, $V_{CTR} = V_S = \frac{R_1 + R_2}{R_2} V_{ref} \Rightarrow 7.5 = \frac{10+60}{60} V_{ref}$

$\therefore V_{ref} = 6.42V$



Application

- Smoke Detector
- Street Light control

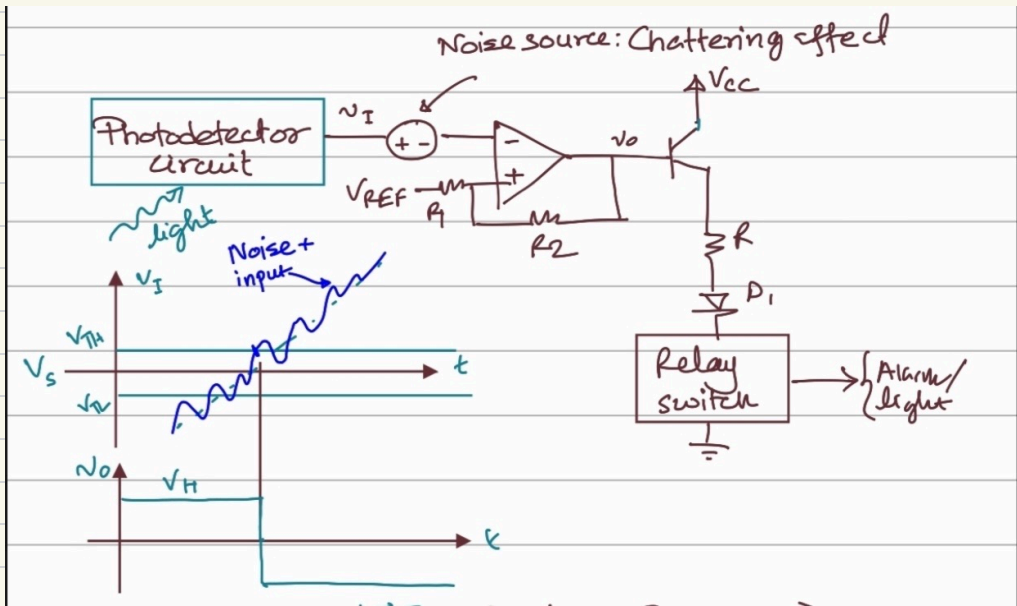


fig: Smoke Detector CKT